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Automated procedural modeling for tunnel inspection and digital twin development

Lucas Bellini Machado [a,*], Bernardo Lopes Poncetti [a], Leandro Silva de Assis [a], Marcos Massao Futai [b]

[a] Escola Politécnica da Universidade de São Paulo, Programa de Pós-graduação em Engenharia Civil, São Paulo, Brazil
[b] Escola Politécnica da Universidade de São Paulo, Departamento de Engenharia de Estruturas e Geotécnica, São Paulo, Brazil

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ABSTRACT

The creation of structured informational models improves understanding of tunnel behavior throughout its service life and enables faster detection of defects, which is critical for hazard assessment and infrastructure safety. Recent studies highlight the importance of automation and 3D reconstruction in overcoming the limitations of manual tunnel inspection and data management. Technologies such as Point Cloud Data (PCD) and advanced modeling techniques have enhanced inspection processes and model generation. However, significant challenges persist in the development of Digital Twins (DT), particularly in the integration of heterogeneous data sources. This paper presents a methodology that combines parametric modeling technics with field data to develop an automated procedural modeling framework for tunnel inspection Digital Twins that integrates heterogeneous data and optimizes lifecycle management. To support this goal, the framework also aims to: (i) automate modeling workflows; (ii) enable the creation of a structured databank; (iii) improve inspection accuracy; and (iv) introduce a new index, the Mesh Decimation Quality (MDQ), to evaluate mesh quality during the decimation and digitization process. The approach was validated through a case study of the Coastal Tunnel in Brazil, for which an automated digital twin was created using PCD from the Faro Focus 3D X330 laser scanner. The MDQ index was applied to evaluate mesh resolution, ensuring model accuracy and controlling computational costs. Metadata from the physical asset was integrated using procedural code scripts. A proposed digital inspection approach, utilizing a procedural inspection grid, successfully identified defects and classified areas, demonstrating the method efficiency and accuracy. This framework highlights the feasibility of digitalizing tunnel inspections and integrating real-world data for improved infrastructure management, with potential for future expansion.

1. Introduction

Tunnels undergo structural degradation over time due environmental conditions, aging, lining excessive loads, and uncertainties in geotechnical properties (Baji et al., 2017; Huang et al., 2017; Montero et al., 2015). Deterioration in the tunnel lining undermines its integrity and may increase the risk of tunnel lining to eventually collapse (Jiang et al., 2019). The inspection of these structures is a crucial process to comprehend how the impact of these factors impairs the structure and to manage maintenance.

Conventional inspection methodologies exhibit known disadvantages, including lower time efficiency and challenges in precisely quantifying and locate defects (Koch et al., 2015; Xue et al., 2022). To improve the inspection practice, several manuals and standards have a specific classification system which classifies the tunnel condition with different approaches and level of deterioration (Poncetti et al., 2025) (Table A.1).

In general, the inspection practices and the condition classification of tunnels rely on visual inspections, with the documentation done in spreadsheets in the field, creating high subjectivity and dependency on the inspector's experience and attention during the defect and condition documentation. Although there are some advances by the creation of manuals and the standardization of the tunnel inspection practice, some improvements are still necessary, specially by adopting the

* Corresponding author.
E-mail addresses: lucasbmachado@usp.br (L.B. Machado), bernardoponcetti@usp.br (B.L. Poncetti), leandroassis@usp.br (L.S. de Assis), futai@usp.br (M.M. Futai).

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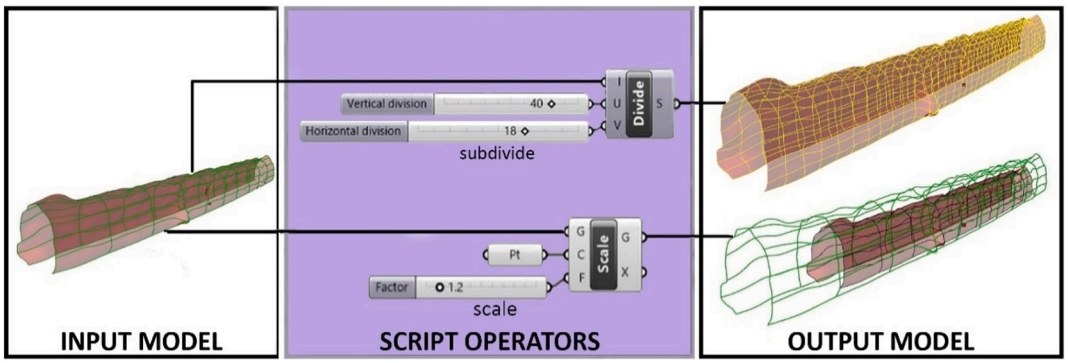


Fig. 1. A procedural script implementation in procedural/parametrical modeling.

technological advances for a better condition assessment, more safety during the inspections and better documentation systems.

Tunnels are critical infrastructures that significantly contribute to the economic and social well-being of populations and play a vital role in the development of transportation networks (Petroutsatou et al., 2021). Regular inspections are essential to monitor the progression of structural pathologies over time, enabling the evaluation of both structural integrity and serviceability (Sandroni & Labiouse, 2011). If left undetected, such deteriorations may lead to severe structural degradation, potentially shortening the tunnel service life, disrupting traffic flow, and compromising user safety. These failures often result in economic losses and require costly remediation measures, while also increasing the risk of failure (McKibbins et al., 2009).

To reduce the problems in the current tunnel inspection procedures, some researches intend to improve the practices by suggesting novel methods for structural condition evaluation (Ai & Yuan, 2018), optimizing the inspection frequency (Ai et al., 2020), or applying technology to the inspection practices, such as using Non-destructive methods (Llanca et al., 2017) and robots (Menendez et al., 2018) to improve the data acquisition, and using Digital Twins (DT) for a better documentation and visualization of the inspection activity.

A DT can be defined as a continuously evolving digital representation of the historical and real-time behavior of a physical asset or process (Dan et al., 2022; Glaessgen & Stargel, 2012). Constructed using real measurements and layers of field data acquisition, digital twins offer significant potential for infrastructure management (Parrott and Warshaw, 2017). However, realizing this potential entails challenges, particularly in managing data heterogeneity. Integrating and synchronizing data from diverse sources is a complex and time-intensive task, which, as noted by Q. Lu et al. (2020) increases the risk of the digital twin becoming outdated. Additionally, while continuous inspection and monitoring practices generate larger datasets that enhance accuracy, they also demand more advanced data management systems. In the context of tunnel inspection, monitoring, and evaluation, digital twins can significantly improve inspection practices and overall infrastructure management.

An iterative virtual model for generating a tunnel DT able to interact with inspection and monitoring database enhances the tunnel evaluation and documentation. This interactive Digital Twin is possible by integrating Point Cloud Data (PCD), parametric modeling, and procedural modeling techniques. PCD enables precise tridimensional representations, as demonstrated in studies such as Zhang et al. (2014) and R. Lu & Brilakis (2019). Furthermore, Sun (2011) presents image acquisition for detecting deformations and defects in circular shield tunnels using mobile laser scanning, focusing on automated DT creation with high detailed geometry and damage integration. To evaluate the geometrical and structural integrity of tunnels, Loprencipe et al. (2018) combined laser scanning, thermographic imaging, and ground-penetrating radar (GPR) with implementation scripts developed specifically for clearance verification. Parametric modeling provides a robust approach for generating flexible geometric models, emphasizing the sequence of mathematical operations and the model construction process, focusing on associative construction rather than on the final outcome (Mafipour et al., 2023; Schultz et al., 2017).

While current inspection practices using laser scanning, photogrammetry, and manual reporting provide valuable data, they often suffer from fragmented workflows, limited automation, and high subjectivity in defect interpretation and quantification (Insa-Iglesias et al., 2021; Stentoumis et al., 2015; Xue et al., 2021). Additionally, these approaches frequently lack seamless integration between data acquisition and model generation, hindering updates and reuse of models throughout the asset lifecycle. This disconnect significantly limits the implementation of Digital Twin methodologies, which depend on continuous synchronization between physical infrastructure and its digital counterpart, particularly in complex environments such as the irregular geometry of the Coastal Tunnel.

Understanding that three-dimensional reconstruction and the automation of structural defect identification and quantification are essential for enhancing inspection quality and meeting infrastructure needs, this paper focuses on implementing parametric modeling and field data extraction to automate modeling processes, improve inspection accuracy, and create a structured databank for the lifecycle management of tunnel digital twins. The objective of this study is to develop an automated procedural modeling framework for tunnel inspection DT, aiming to integrate heterogeneous data sources and support lifecycle management through improved digital inspection and classification capabilities.

The workflow integrates field data obtained from detailed inspection reports and structural scans as inputs for a Grasshopper 1.0.0007 script. It involves surface creation, adaptive subdivision into an inspection grid, and automated classification of areas based on digital inspection data. Metadata containing relevant information is then assigned to areas of interest.

The methodology was validated through a real-world application in the Coastal Tunnel in Brazil. This approach enabled virtual inspections and established a virtualization environment that seamlessly integrates real data with detailed modeling. It allows the full execution of the inspection and classification campaign digitally within the scanned area, enabling updates and refinements based on acquired data and developed programming. By reducing subjectivity and incorporating Industry 4.0 methodologies, this work represents a significant advancement in tunnel management practices.

2. Parametric/procedural modeling fundamentals

The main requirement to create a DT is the direct connection between the model and the infrastructure. Automation, a detailed geometry and metadata insertion are essential for bridging the real and digital environments. In this context, the parametric and procedural modeling allows creating a realistic 3D model able to adapt to the data input and visualization to being adaptable to different applications of the DT in the

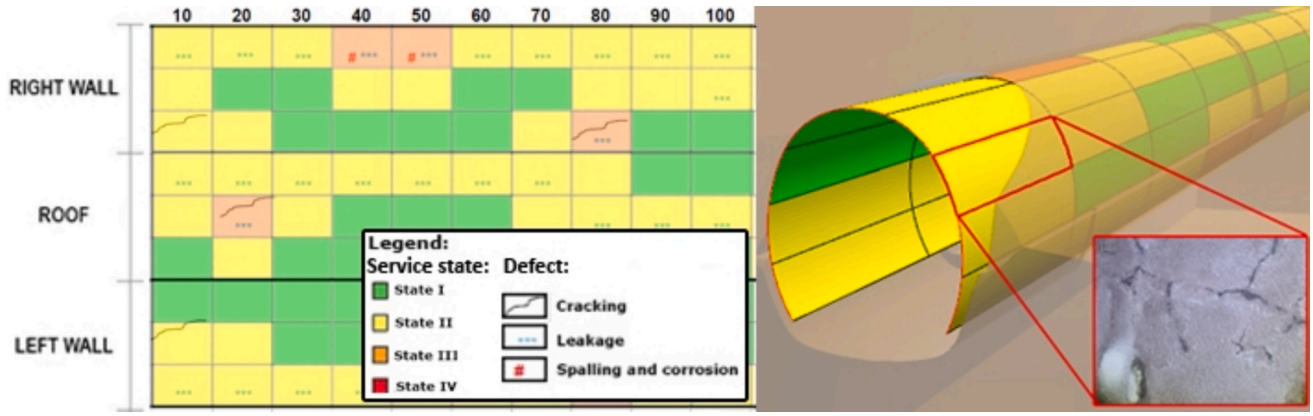


Fig. 2. Inspection methodology model.

tunnel management activities.

Parametric modeling can be an important tool for integration and for creating a virtualization environment for the structure. For example, Sung et al. (2020) utilize visual programming and parametric modeling to quickly calculate the layout of a construction project in the field of civil engineering. Erharter et al. (2023) use this tool for parametric modeling and BIM model generation for the geology of Tunnel Angath. Utilizing advanced modeling techniques in the case study, such as parametric programming, allowed visualizing the irregular excavation geometry due to the modeling flexibility, meeting the demand for updating current methodologies for representing such structures.

The process uses mathematical standardizations with adjustable parameters to create geometry, which can be modified to enhance the accuracy or suitability of the model to field data or design adjustments (Fig. 1). The Grasshopper 1.0.0007 parametric modeling tool operates as a graphical interface for algorithm editing. It is a plug-in for the three-dimensional modeling software Rhinoceros 7 3D, enabling to create parametric 3D models and to generate complex geometries iteratively and flexibly. The visual algorithms consist of interconnected components; each one performing a specific transformation or operation to adjust the components and modify the resulting geometry in an automated and flexible manner.

Fig. 1 illustrates the process of parametric modeling using graphical algorithms in the Grasshopper language. This approach is based on the interconnection of input data through a series of modeling operators. In the provided example, the parametric graphical operators are associated in parallel; the scale of the grid structure on the tunnel surface is increased by 20 % from the designated point and the division is also increased. The resulting new mesh, after implementing this process change, is made available at two different outputs, enabling a sequence of programming modifications and operator associations.

Thus, with minor adjustments, the entire model can be updated. This type of programming demands significant processing power, as more complex modeling structures require a complete recompilation of the framework when changes are made to the initial systems. Consequently, programming optimization becomes essential. Optimization adjustments can be implemented without the need for an extensive series of associations. While this may result in some loss of control over the structure properties, it can significantly reduce computational costs.

3. Methodology

The proposed methodology is divided into two parts. First, it explains the inspection procedure, which serves as the input for generating the data input for digital twin. Second, it details the proposed parametric implementation in Grasshopper code, clarifying each step involved in constructing the automation process. This is crucial because each parameter and stage can be adapted flexibly to meet specific demands.

3.1. Inspection data generation

The standardization of inspection practices has a global scope, with various companies responsible for infrastructure while administrative associations pursue an ideal methodology (ADIF, 2021; CeTU, 2015; FHWA, 2015; Yuan et al., 2012). Commonly applied practice involves categorizing areas within the structure using defect-sensitive classification. Response data plays a crucial role in guiding and enhancing the decision-making system. Machado & Futai (2024) present a methodology to create a virtual representation for the inspection data (Fig. 2).

The inspection process involves creating a grid that divides the tunnel surface into zones: the roof, the right wall, and the left wall. These zones are further divided into three longitudinal areas, each spanning 10 m in the transversal direction. This grid is used to locate structural defects and classify them into four service states, in which State I represents an intact condition and State IV indicates severe damage. Metadata information is then inserted into specific zones based on field data (see Machado & Futai (2024) for more details). However, some challenges are associated with this methodology. Tunnels often have irregular geometries, making it difficult to define specific locations and accurately register field inspection notes. To address this, Machado & Futai (2024) proposed using a zone of influence for classification: if a State I defect is located between two State II zones, it will be registered as State II, since the low level of detail in the model prevents the precise localization of pathologies. Although this approach does not fully solve the problem, it enhances safety by adopting a conservative approximation. The advanced modernization brought by Industry 4.0 can help overcome these challenges by enabling digital inspections, refining the inspection grid, creating more detailed models, and automatically inserting metadata and classifications.

The modernization of inspection practices in tunnel infrastructures is based on generating more detailed data about the structure, thus creating a database to optimize the decision-making process, and managing the structure lifecycle. The methodological framework was based on infrastructure inspection, laser scanning of underground infrastructures, and modeling associated with this data. These foundations aim to meet the requirements for the virtual twinning of the tunnel.

3.2. Tunnel automated implementation

Fig. 3 presents the proposed framework for automated generation implementation of the inspection layer and lifecycle management for creating a Tunnel Inspection DT. The methodology uses Point Cloud Data (PCD) generated by Terrestrial Laser Scanning (TLS) and sensitivity inspection results as input. In the initial stage, the point cloud undergoes a cleaning and decimation process to remove redundant and excessive points, resulting in a more manageable and workable dataset. The next step involves mesh generation and auxiliary cutting and fitting

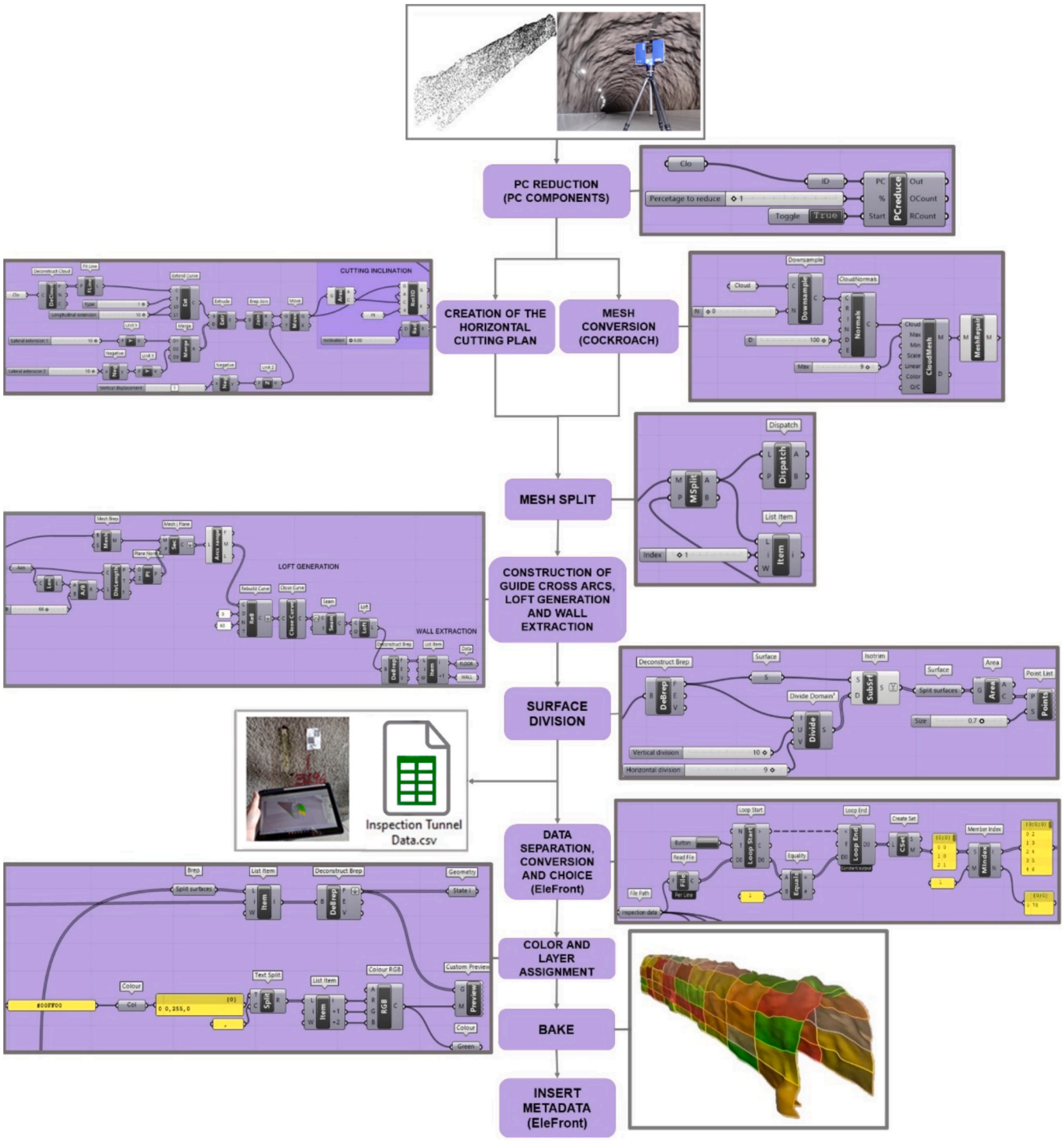


Fig. 3. Automated Inspection Tunnel DT procedural implementation framework.

geometries. Subsequently, the mesh is used for constructing guide arcs and creating the three-dimensional grid. The selection of the generation arcs is used as a cross-section of the surface. Consequently, classification data of the areas are introduced, categorizing the grid areas into four degradation states by layer and color. In the final step, the EleFront 5.1.8 application is employed to assign metadata derived from sensitive inspections to the necessary grid areas, including inspection notes, photos from online databases, and information relevant in the implementation environment to associate with the automated Inspection Digital Twin.

Fig. 3 presents the data inputs, results, and graphical programming clusters used for each phase of the methodology. Each specific step will be described and explained in detail in the case study to demonstrate the development of the methodology. Within the programming, there are variable values that can be adapted and adjusted to meet specific cases and demands. Examples include the inclination of the horizontal cutting plane, adjustments for mesh creation to enhance the quality of the PCD, and the range of guide arcs for surface generation, among others. These aspects are further discussed in the case study, emphasizing the flexibility and adaptability of the approach to accommodate varying demands and field data quality. This adaptability is one of the most significant advantages of using a parametric approach.

Between the mesh split and the construction of guide cross arcs, the

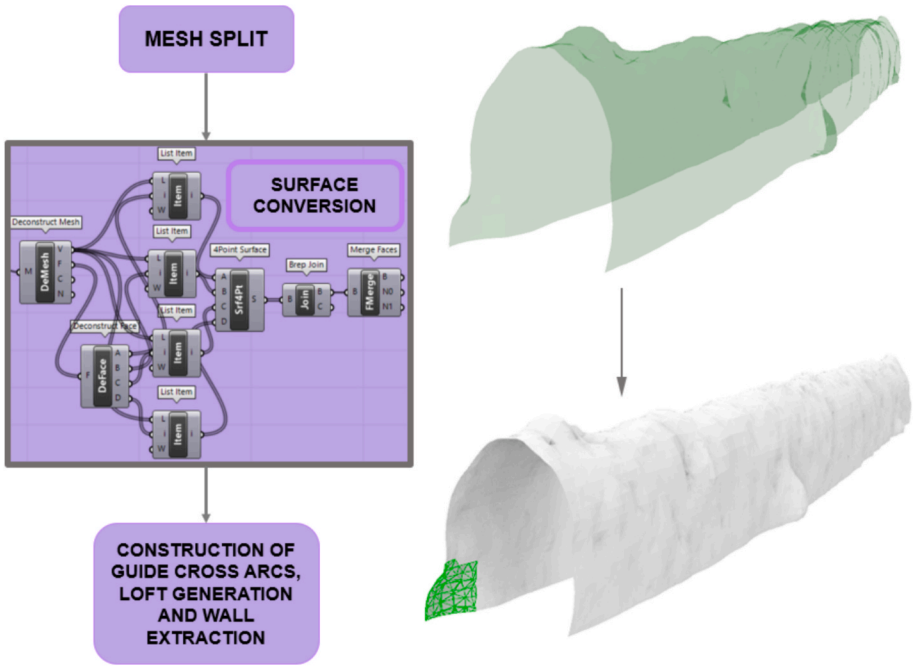


Fig. 4. Convert mesh to individual surfaces to trim and adjust the generated geometry.

mesh can optionally be converted into a surface for minor adjustments (Fig. 4). This implementation transforms the unitary mesh into multiple surfaces generated by connecting points. The conversion facilitates localized mesh cutting and can be adapted for features that require surface input. However, it is a computationally expensive process and may not be necessary for small adjustments. As a result, surface conversion can be considered an optional step, although it may enhance the methodology in certain cases.

To construct the mesh from the PCD and generate a surface, the open-source extension Cockroach 0.01 (Vestartas & Settimi, 2020) for Grasshopper was utilized. This extension employs Poisson Surface Reconstruction (Eq. (1)) to reconstruct the surface. The "CloudMesh" operator is used to apply the Poisson equation, solving a target mesh with known topology and vertex coordinates. To determine the vertex coordinates, the Poisson equation requires a discrete guidance vector field for each of the three coordinates, whose divergence is computed at the vertices of the parameterization mesh (Yu et al., 2024).

$$\nabla^2\chi = \nabla.\vec{V} \tag{1}$$

where χ is the scalar function that represent the surface, $\vec{V}$ is the guidance vector field, ∇^2 is the Laplacian operator (Eq. (2)), and $\nabla.\vec{V}$ is the divergence of $\vec{V} = (V_x, V_y, V_z)$ (Eq. (3)).

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2} \tag{2}$$

$$\nabla.\vec{V} = \frac{\partial V_x}{\partial x} + \frac{\partial V_y}{\partial y} + \frac{\partial V_z}{\partial z} \tag{3}$$

Mesh decimation is a technique used to optimize and simplify the underlying shapes of a model by reducing the number of vertices, edges, and, consequently, surfaces that compose the polygons. In modeling, mesh decimation involves simplifying the level of detail/development (LoD) of a mesh without substantially altering its overall shape. This process reduces the size of model files and decreases computational demands, making it a valuable tool for optimizing.

4. Results and discussion

This section is devided into two main subsections. The first focuses on the validation process for the methodology, whereby a generic railway tunnel PCD was used to test the developed parametric model script and to calibrate its parameters. The second subsection involves applying the full methodology, including the inspection campaign data and the Grasshopper framework, to a real highway tunnel in Brazil.

4.1. Validation process

The validation process develops the proposed methodology and calibrates the parameters for tunnel field data infrastructure. The process involves conducting tests and adjustments in the graphical code programming using a railway tunnel point cloud. The implementation of procedural modeling to automate the reconstruction of tunnel models is infrequently addressed in the literature. Therefore, the proposed framework and tests provide valuable insights into the application of this approach to such infrastructure.

4.1.1. PCD processing and optimization

Point cloud cleaning and segmentation are crucial steps in the result processing of laser scanning or other 3D mapping technologies data. These steps aim to eliminate inconsistencies, noise, and unwanted regions from the point cloud, resulting in more precise and usable data. Filtering based on density, distance, duplicates, and outlier removal are common techniques employed for this purpose (Oytun & Atasoy, 2022).

Distance filtering with outlier elimination was performed by defining the Region of Interest (ROI), and external points were removed using the Crop function facilitated by the Crockroach plugin, which allows point information editing without the need of conversion into editable elements. Given the complexity of directly editing point files, the point cloud was trimmed using editing tools within Rhinoceros 7 without requiring conversion into editable elements.

Mesh decimation was performed to improve manageability by applying density filtering to the raw acquired point cloud file using the PC reduce function, implemented by the open-source Point Cloud Components (PCC) 1.0 plugin. A new quality index, Mesh Decimation Quality (MDQ), is proposed to evaluate mesh quality after the decima-

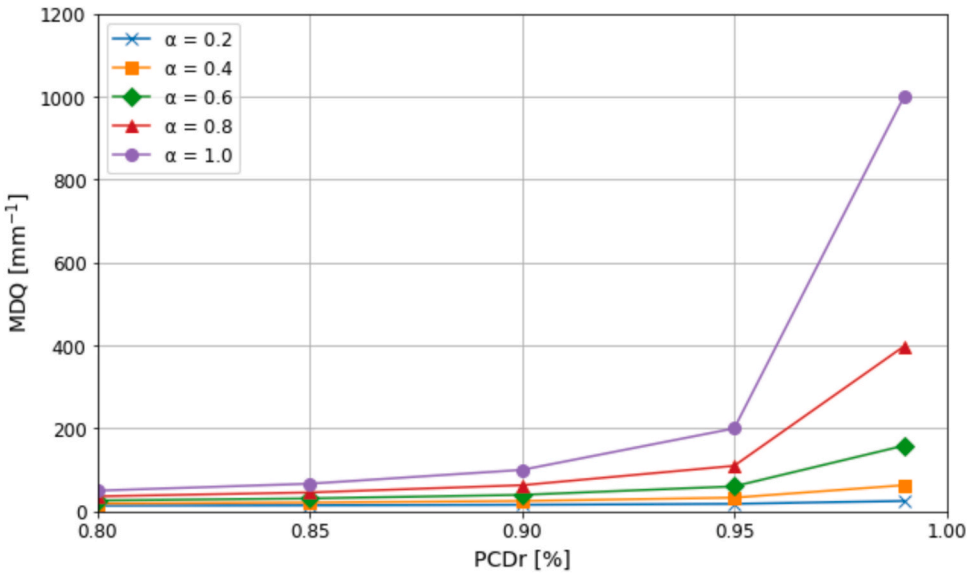


Fig. 5. MDQ exponent α calibration to minimize the PCD reduction influence.

Table 1
Decimation quality metrics analysis.

PCD Reduction [%]	Avg. Dist [mm]	Hausdorff Dist. [mm]	RMSE [mm]	Avg. FND [rad]	MDQ [mm^{-1}]
20 %	6.533	508.680	25.037	1.240	9.703
40 %	7.798	513.118	26.026	1.440	10.173
60 %	16.760	519.110	34.213	1.314	11.224

tion process (Eq. (4)).

$$MDQ = \frac{OAR}{AD \times (1 - PCDr)^{\alpha}} \tag{4}$$

where OAR represents the overlayed area relation, quantifying the relationship between the reduced mesh area and the raw mesh area; AD denotes the average distance between the vertices of the decimated mesh and the closest point in the raw point cloud data (PCD); and PCDr refers to the reduced PCD. To minimize the influence of PCDr on the index, an exponent α was calibrated for this purpose (Fig. 5). The value of α = 0.2 was adopted to calculate the MDQ without distorting the results due to the PCDr factor. The index was designed to prioritize point reduction while preserving mesh quality, as measured by the Overlayed Area and Average Distance (Fig. 5).

Different metrics for mesh decimation were compared to evaluate and validate the applicability of the MDQ approach (Table 1 and Fig. 6). The Hausdorff Distance is a commonly used metric that measures the maximum of the minimum distances between two sets of points (e.g., a mesh and a point cloud, or an original and a decimated mesh), but it is highly sensitive to outliers (Kobbelt et al., 1998). Root Mean Square Error (RMSE) represents the quadratic mean of the positional differences between corresponding vertices in the decimated and original meshes; because it squares the errors before averaging, RMSE tends to amplify larger deviations. Face Normal Deviation (FND) quantifies the angular difference between the normals of corresponding faces in the original and decimated meshes. This metric is used to assess distortions in shape and surface curvature introduced by decimation; however, it requires accurate face-to-face correspondence, which is typically evaluated using the Overlayed Area Relation (OAR).

To compare the behavior of the MDQ with other commonly used

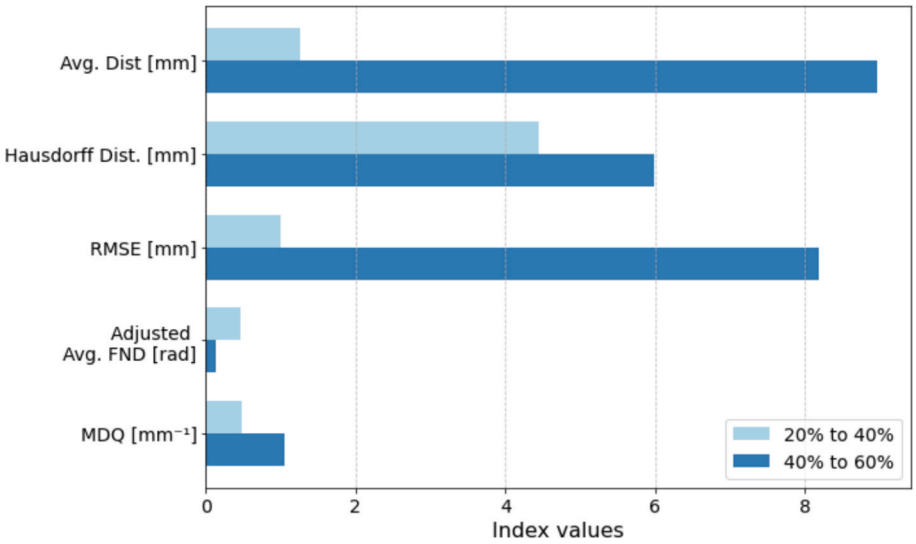
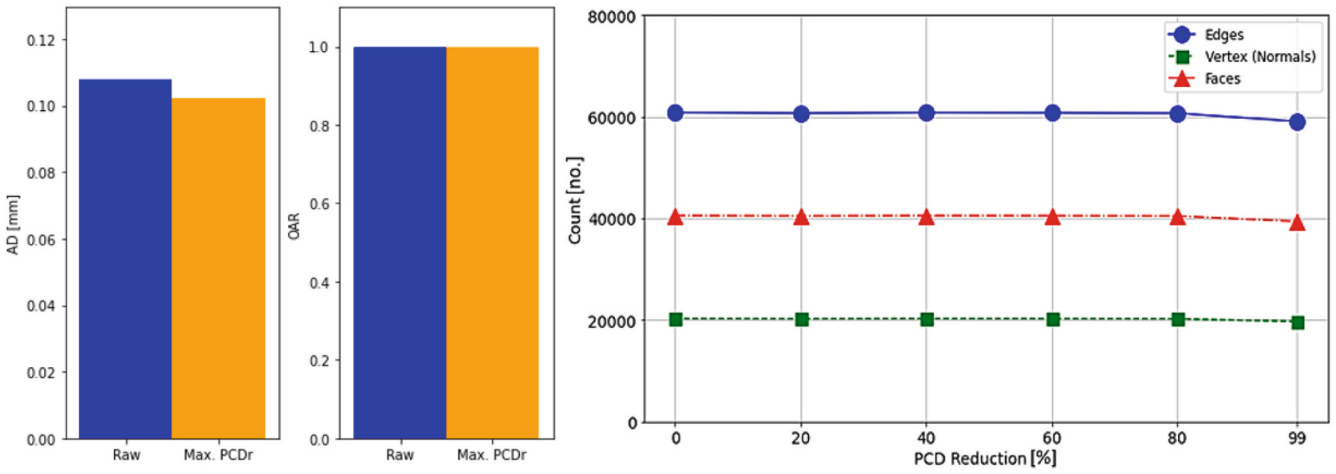
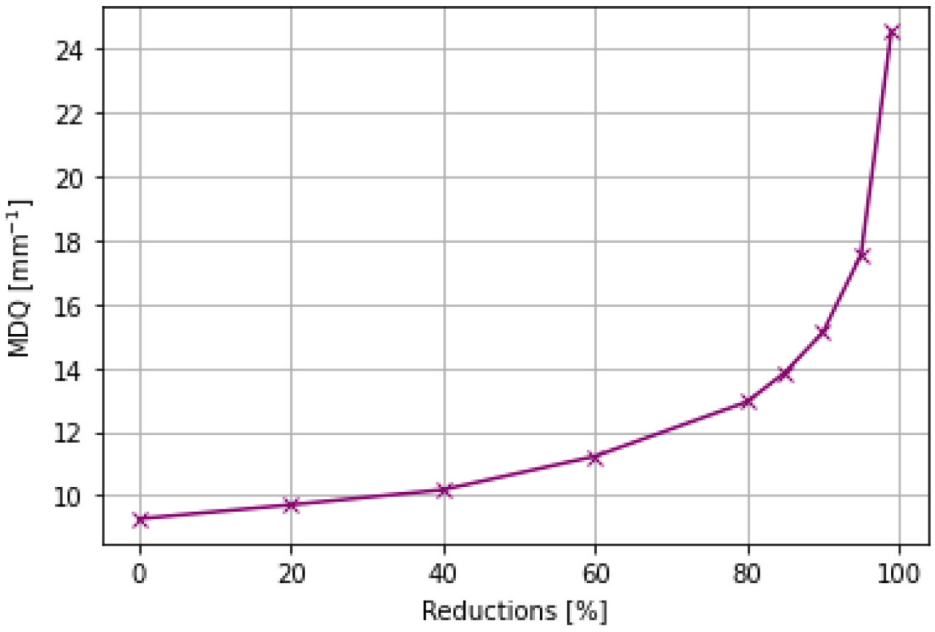


Fig. 6. Quality mesh indices variation with PCD reduction.



(a)



(b)

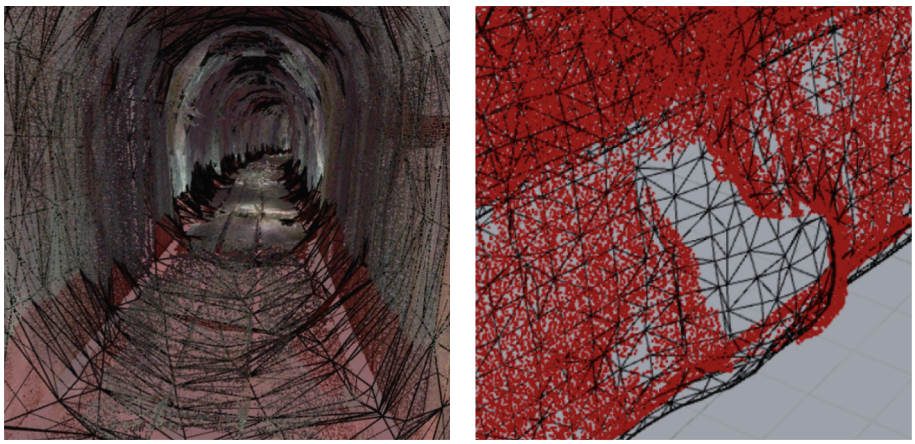
Fig. 7. Decimation tests for tunnel PCD: (a) mesh shape and density parameters; (b) MDQ factor.

decimation indices, a tunnel point cloud (PCD) sample was selected, and three consecutive decimations were applied in 20 % reduction steps to generate the meshes and compute the metrics. Table 1 presents the metric values for each reduction level, while Fig. 7 illustrates their variation across the steps. The Hausdorff Distance shows relatively high values, as it reflects the maximum point-to-point distance and is strongly influenced by outliers. This effect becomes clearer when compared to the Avg. Distance, which provides a stabler representation. Similarly, RMSE values are elevated due to the squaring of errors, which amplifies larger deviations.

For the Average Face Normal Deviation (Avg. FND), the angular difference decreases in the final reduction step. This metric is highly sensitive to mesh restructuring and, in some cases, the orientation of face normal in the decimated mesh becomes more aligned with the original, reducing the deviation. While standard decimation quality indices show higher values as the difference between the raw and decimated meshes increases, the proposed MDQ shows the opposite trend: higher values indicate a better match between the meshes. These contrasting behaviors support the MDQ approach, which takes into account both spatial distance and mesh adjustment, with reduced sensitivity to outliers, in contrast to RMSE, as shown in Fig. 6.

The MDQ equation incorporates the OAR to capture face-to-face correspondence and the AD to account for vertex spacing. Between 40 % and 60 % decimation, the OAR increases by approximately 0.025, reflecting greater mesh alignment and confirming the MDQ sensitivity to structural adjustments, similarly to the trend observed in Avg. FND. Meanwhile, the AD continues to increase, indicating a growing spatial gap between corresponding points.

Using the MDQ, Fig. 7.a shows that AD and AOR are similar in both the maximum test reduction and the raw data. Additionally, the number of elements composing the density remains very similar, even though the points are reduced from over 1 million to around 12 thousand, and



(a) (b)

Fig. 8. Implementation test and validation in railway tunnel: (a) Floor-wall connection; (b) Approximation in areas with low point cloud density.

the file size drops to under MB unity. This suggests that it is possible to drastically reduce the number of points without significantly compromising mesh quality. Furthermore, the MDQ increases with the reduction because, within the evaluated range, the shape quality is maintained as shown in Fig. 7.b.

4.1.2. Mesh generation

The mesh generation process primarily depends on a PCD suitable for the developed methodology. Regions with redundant points or high density significantly increase processing time, as described in the previous section. However, improper cleaning, particularly concerning outliers, can result in incorrect or inadequate meshes in terms of LoD. To improve the adaptation of the process to the specific characteristics of the real tunnel case, a railway unlined simple tunnel point cloud was tested (Fig. 8).

Fig. 8 illustrates the application of the methodology to a railway tunnel during validation tests prior to practical implementation. The mesh construction, guided by the point cloud, is achieved using Poisson Surface Reconstruction through the open-source plugin Cockroach 0.01, which provides point cloud commands and post-processing capabilities within the Grasshopper graphical programming environment. The connection between the Normals and CloudMesh functions generates the surface from the field data information (Fig. 3).

The absence of guide points at the connection between the walls and the floor of the structure presents challenges for mesh generation in these specific areas (Fig. 8.a). However, it is possible to minimize errors by adjusting the properties of the parametric model, reducing decimation, and using raw data or modifying mesh generation parameters, although these processes increase computational cost. Areas with low point cloud density or missing data due occlusion effect, but without significant changes in directional continuity are filled by approximation, representing a substantial gain in the methodology (Fig. 8.b).

The trimming process for the mesh differs between longitudinal and transverse cuts (Fig. 9.a). Longitudinal cuts use an adjustable cutting plane, with its extensions and inclination relative to the axis defined as critical for the structure inclination. The script generates a median line from the point cloud and extends it to create a cutting plane. The lateral extension, height, and angle of this plane can be calibrated. Transverse cuts are defined in the next stage using arcs from the inspection grid, generated based on the mesh. Only arcs delimiting the walls are selected for trimming, while the first and last arcs serve solely as boundaries reference for the procedure.

The trim longitudinal code is presented and adjusted using scrolls in the Grasshopper programming environment (Fig. 9.b). These adjustments are made to ensure effective intersection between the point cloud and the created surface. The longitudinal mesh trim involves creating a horizontal plane, which uses point cloud data to define the components for adjusting the cutting process, including plane inclination and the length of the cutting tool. The transverse trim (Fig. 9.c) involves three stages: The first step creates the arcs using the generated mesh, with the user defining the extension and boundary of the set of arcs limits. The second step closes the arcs and generates the loft connecting them.

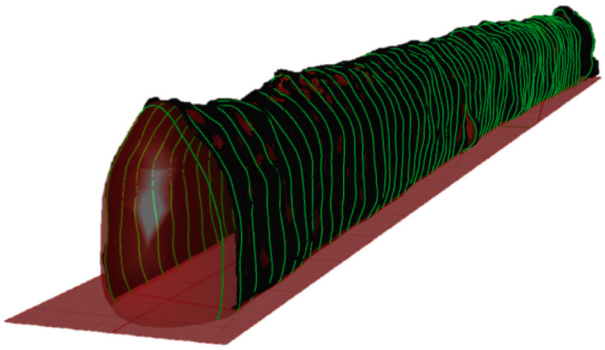
Procedural modeling provides adjustable tools to meet specific modeling preferences or requirements. The transverse and longitudinal cutting algorithms adjust automatically, generating a model with minimal manual tuning. The loft creation function connects the transverse arches, producing separate meshes defined by the arches to generate the grid areas. A significant advantage of this approach lies in the flexibility to create the layers that compose the process. Fig. 10 illustrates this adaptability in the subsequent process of splitting areas for the inspection grid, allowing it to adjust to different patterns and methodologies used by companies conducting integrity analyses.

4.1.3. Metadata information attribution for model database

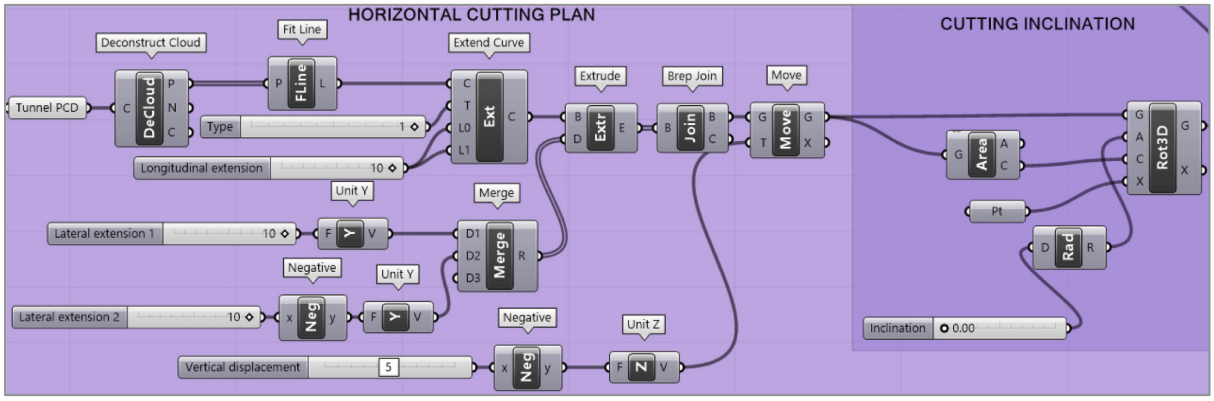
In the final stage of the automated creation of the digital twin, the process of inserting metadata into the generated three-dimensional grid is divided into two steps. First, data from the structure inspection identify structural defects such as leaks, cracks, peeling, and others, and classify the grid areas (Fig. 2). Next, a.csv file is created, containing numbers corresponding to each classified area of the grid. This file is then integrated into the algorithm, which separates the state of each area into layers and automatically applies colors (Fig. 11).

Fig. 11 illustrates the sequence of area classification insertions into the model from the generated.csv file. The simplified file format enables easy integration with new technologies. This methodology ensures flexibility in inspection input, once state classifications are assigned and a sorted data file is generated, the data-driven digital twin can be created automatically.

The open-source EleFront 5.1.8 plugin was used for metadata insertion into geometry in Rhinoceros 7 and Grasshopper 1.0.0007 via object attribute functions (Fig. 12). Metadata, stored as attributes, are integrated into the Digital Twin using a standardized programming dictionary, enabling queries, sorting, and filtering. This facilitates the organization of large models and creates a key-value-based NoSQL database. The methodology also links to online image repositories, allowing the inclusion of inspection photos, maintenance records, reports, geometric data, and other relevant information.



(a)



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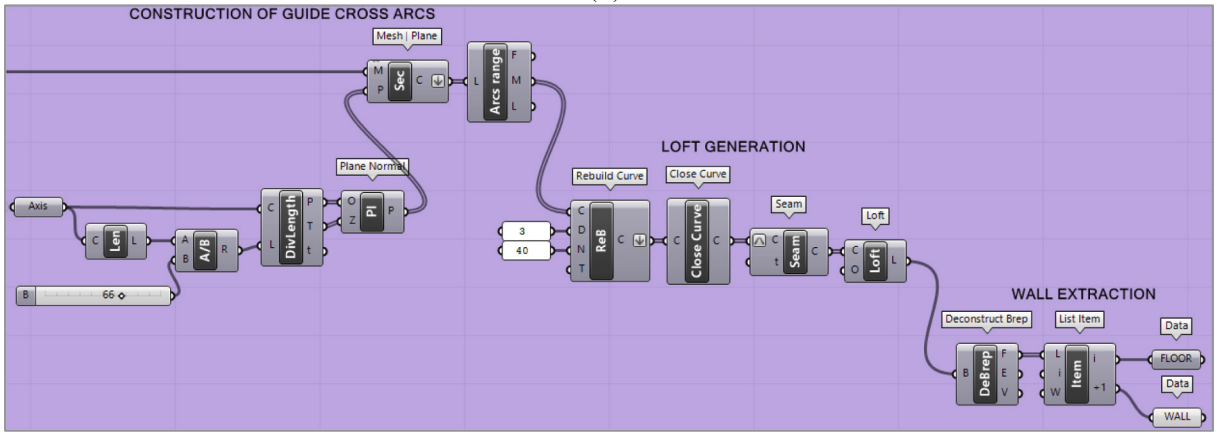


(c)

Fig. 9. Mesh delimitation adjustment for parametric modeling: (a) Representation of the tunnel model in Rhinoceros 7; (b) Implementation for longitudinal planes; c) Arcs for transverse trim, loft generation and wall extraction.

4.2. Coastal tunnel application

The automated construction for inspection digital twin for the structure was implemented in a real highway tunnel (Coastal Tunnel). The tunnel awaits subsequent road connection works before it can begin operations (Fig. 13). Located near the shore coastline in the southeastern region of Brazil, the Coastal Tunnel plays crucial economic and social roles, respectively by facilitating the transportation of materials and for connecting important tourist destinations in the country.

The decision about the specific tunnel for the framework application is primarily due to the irregular geometry resulting from the shotcrete lining of the design. The conventional virtualization process would rely on numerous approximations, neglecting potential construction issues or overbreak zones. Using PCD to feed the procedural programming results in a more detailed indoor mapping and can connect structural information to the model, creating a metadata-connected replica and database for the infrastructure.

The proposed data acquisition strategy and modeling framework are designed to be adaptable and may require minor adjustments for application in tunnels with different geometries or environmental conditions, particularly in cases where point cloud coverage is limited or occlusion areas are more prominent. Tunnel surface geometry is often influenced by the type of lining material. Based on the list of common tunnel linings provided by FHWA (2015), it can be inferred that materials such as steel, precast concrete, or masonry tend to produce regular surfaces, while shotcrete, unlined, or timber-lined tunnels result in more irregular geometries and occlusion issues. Procedural modeling techniques (Fig. 8.b) help address gaps in PCD coverage; however, scanning

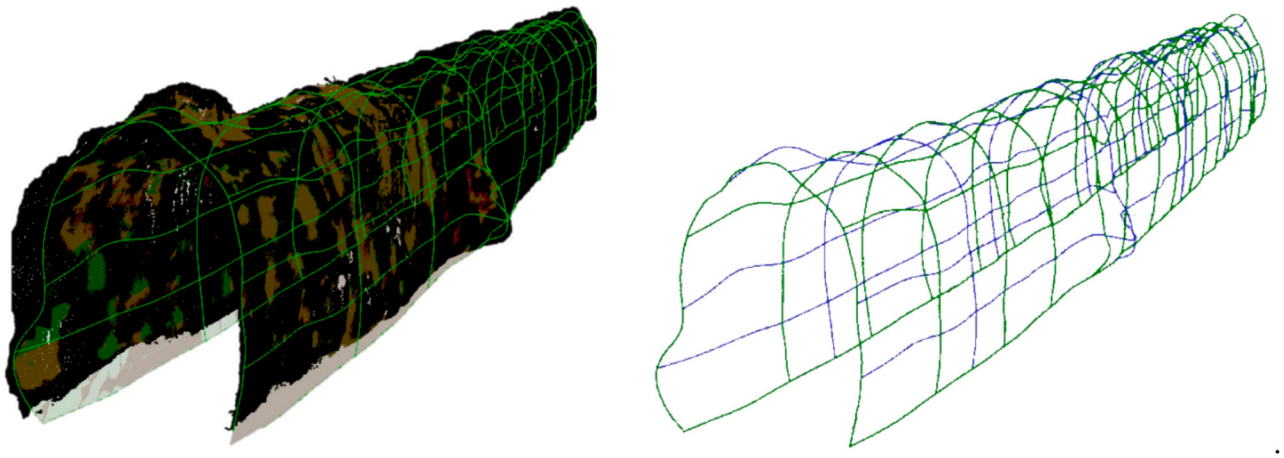


Fig. 10. Grid division process and analysis area creation.

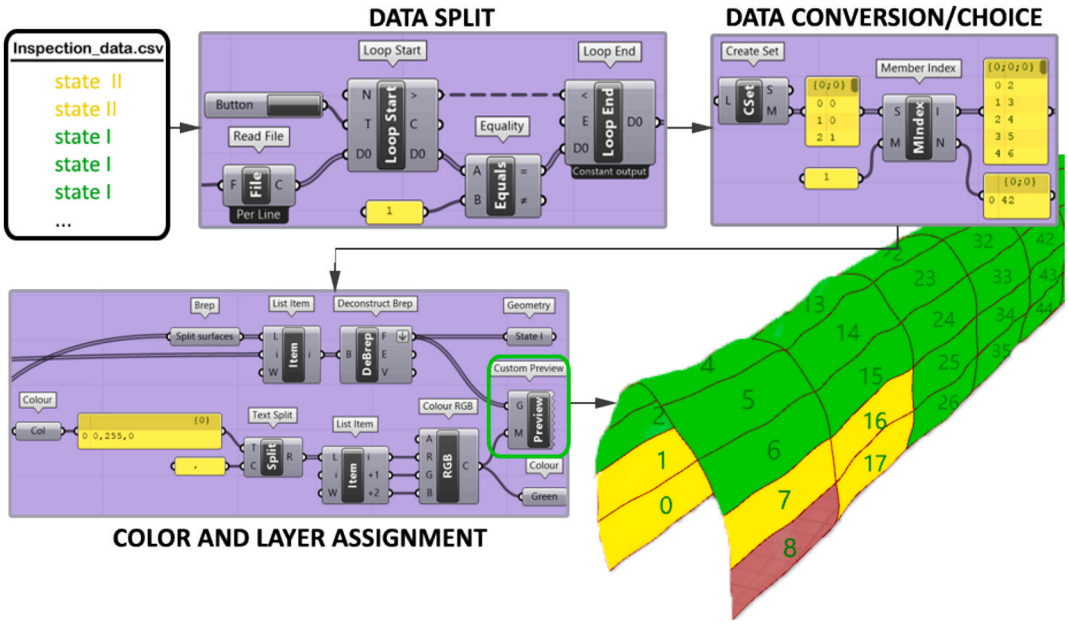


Fig. 11. Inspection script classification and metadata assignment.

strategies may need to be refined by reducing scan intervals or increasing resolution in complex environments.

In scenarios where TLS is impractical, such as in hazardous or hard-to-reach environments like active mines or submerged hydraulic tunnels, alternative acquisition methods using LiDAR-equipped unmanned aerial vehicles (UAVs) or remotely operated vehicles (ROVs) can serve as effective replace (Kang et al., 2025; Wang et al., 2025).

4.2.1. Field data acquisition (Sensitive inspection and Scanning)

The PCD acquisition was performed using the Faro Focus 3D X330 laser scanning equipment in setup configuration resolution 1/5 and quality 4x, for its long-distance measurement capabilities (Fig. 14). The scanning was conducted at approximately 15 m from the portal, covering a sweep of about 60 m in a single reading. The wait time for acquiring the scan data and capturing colors through images is approximately 10 min, of which 6 min are allocated for creating the point cloud and 4 min for color capture. This acquisition process results in a point cloud density of approximately 26.5 million points.

The laser scanner captures point data with varying densities depending on the distance from the object and the angle of incidence. As the distance increases, both resolution and precision tend to decrease, which may affect the quality and accuracy of the field data acquisition. Another source of bias involves occlusion areas, where the scanner cannot capture data due to physical obstacles or complex geometries, resulting in gaps in the point cloud, as illustrated in Fig. 8.b.

The Faro Focus 3D X330 scanner has been previously validated in unlined tunnel environments for modeling purposes (Cacciari and Futai, 2017). This validation supports the reliability of the acquired data in this study, despite the natural limitations of the scanner, such as resolution variability and occlusion effects.

A sensitive inspection was conducted using a generic three-dimensional model created in Rhinoceros 7 for classification and direct acquisition of digital data. A tablet was employed to access the main hardware via internet and to acquire data directly into the main storage (Fig. 15). The data on the geometry was directly recorded using the Rhinoceros 7 3D user text attribute function. The data was conventionally recorded in a 2D grid for a certain level of redundancy in the tunnel inspection process. The limitations of internet connectivity in a tunnel can be mitigated by combining Low Earth Orbit (LEO) satellite internet and Long Range (LoRa) technologies.

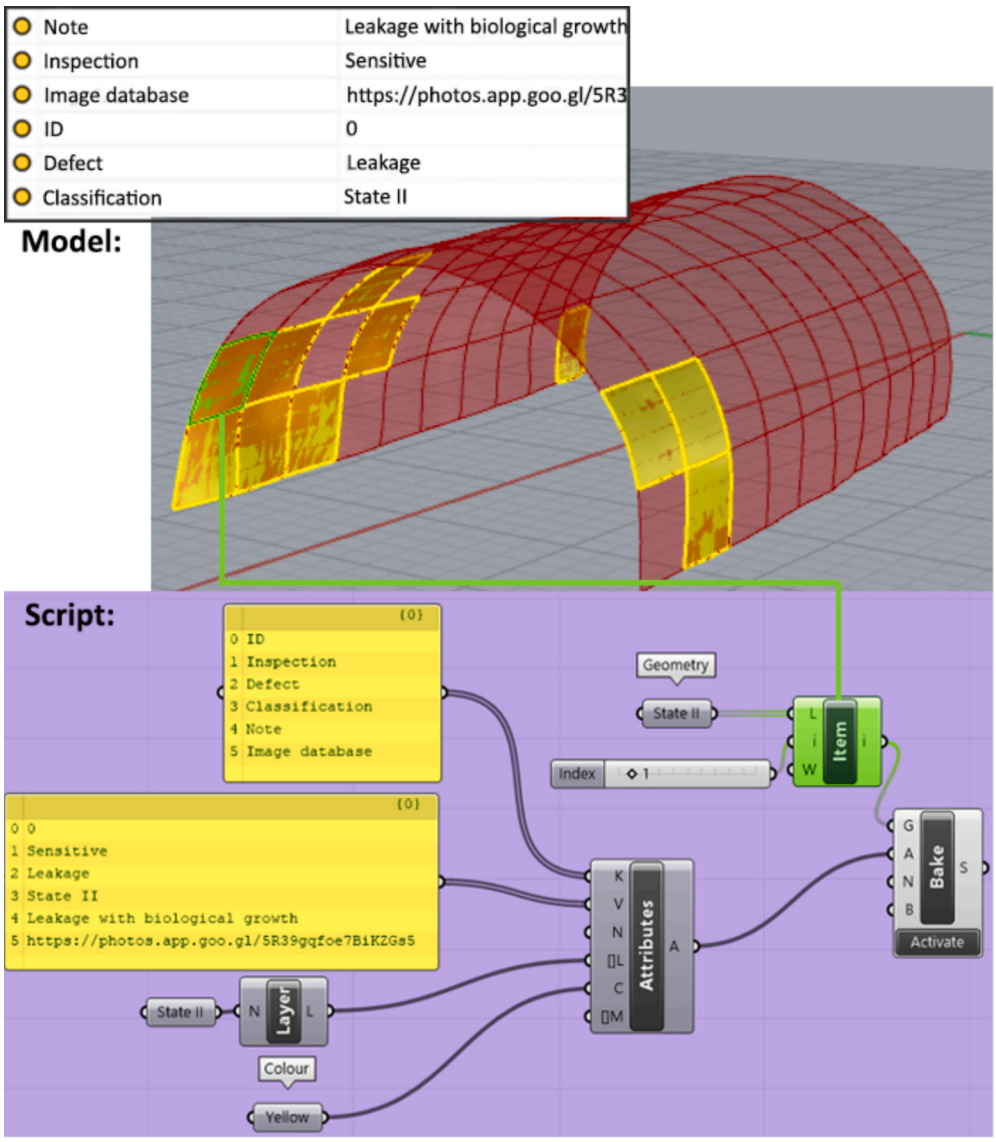


Fig. 12. Inspection classification and metadata input in script code.

Fig. 13. Coastal Tunnel case study for framework application.

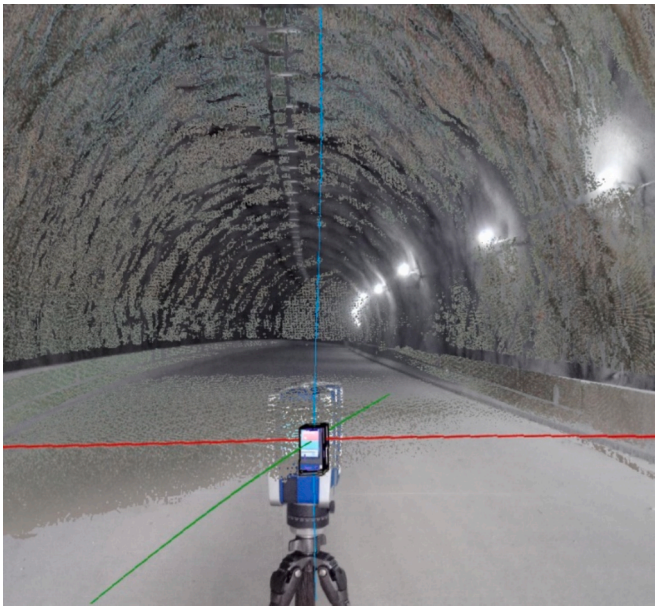


Fig. 14. Overlay scanning and collected data in the Coastal Tunnel.

4.2.2. Parametric modeling with field data for the Coastal Tunnel

A single tunnel scan was employed as a systematic sampling approach. This decision aimed to evaluate the extent of mesh generation from a single scan and to calibrate the resolution of the data acquisition process. The selected PCD was collected near the tunnel portal, a region characterized by significant brightness variation to test color capture. Additionally, this area benefits from a stable internet connection and contains the highest concentration of diverse defect types (e.g., cracking, spalling, leakage, and construction-related issues).

The PCD was compiled and exported to Rhinoceros 7. The process of removing outliers and cleaning from the defined ROI is carried out by cutting using the Volvox 0.3.0.0 Grasshopper plugin. The point cloud enters as a set of points, and separates these points in individual value results in high processing demand; each of the millions of points has its coordinates. Therefore, for removing the region of interest from RAW data, the use of specific PCD cutting tools is recommended. In subsequent steps, localized geometry adjustments can be made using the surface conversion methodology presented in Fig. 4.

The filter process started with a dataset of over 26 million points acquired from raw TLS field data. The reduction methodology was applied to optimize modeling while ensuring no loss of resolution during surface creation, as shown in Fig. 7. Therefore, automatic reconstruction programming was applied to the treated point cloud, constructing the structure with procedural modeling. In a subsequent process, the transverse cutting arcs were selected for transversal trim.

Fig. 16 shows the properties of the final adopted mesh after the decimation process. In Fig. 16.a, the significant reduction in mesh resolution is evident, aimed at improving workability and at reducing computational costs. Fig. 16.b displays the final mesh and the calculated indexes for the process. The MDQ increased, while the AD decreased significantly, and the OAR remained relatively stable. The mesh fits well for creating the inspection digital grid, validating the proposed MDQ index.

4.2.3. Digital inspection and parametrical implementation

A key aspect of digital twin construction is the Fit Line (Fline) function (Fig. 8.b), which generates a guide axis based on point cloud averages. This function depends on PCD point density, and on the experimental design, Fline required three-dimensional position adjustments using the 3D Rotate function. The same adjustment was applied to plane cut inclination for longitudinal cutting and to guide arc creation, which defines grid classification areas. In the Coastal Tunnel, TLS-generated point clouds ensure uniform data distribution, enabling Fline to construct an accurate axis for the methodology.

The methodology developed combines procedural modeling programming with colored point cloud data to classify structure degradation levels virtually. It utilizes the ROI of the point cloud extracted by scanning and generates grid areas based on it, facilitating accurate classification according to company standards (Fig. 17). The process greatly relies on the structure luminosity since the TLS captures data from images.

Fig. 17 presents a result from the next phase of programming, which involves extracting the tunnel assessment grid curve. This methodology enables the full execution of the inspection and classification campaign digitally within the scanned area, allowing updates and refinements based on acquired data and developed programming (Fig. 17.b). Notably, excess brightness near the portals prevents accurate color extraction from the PCD acquisition. In this case, an on-site inspection is

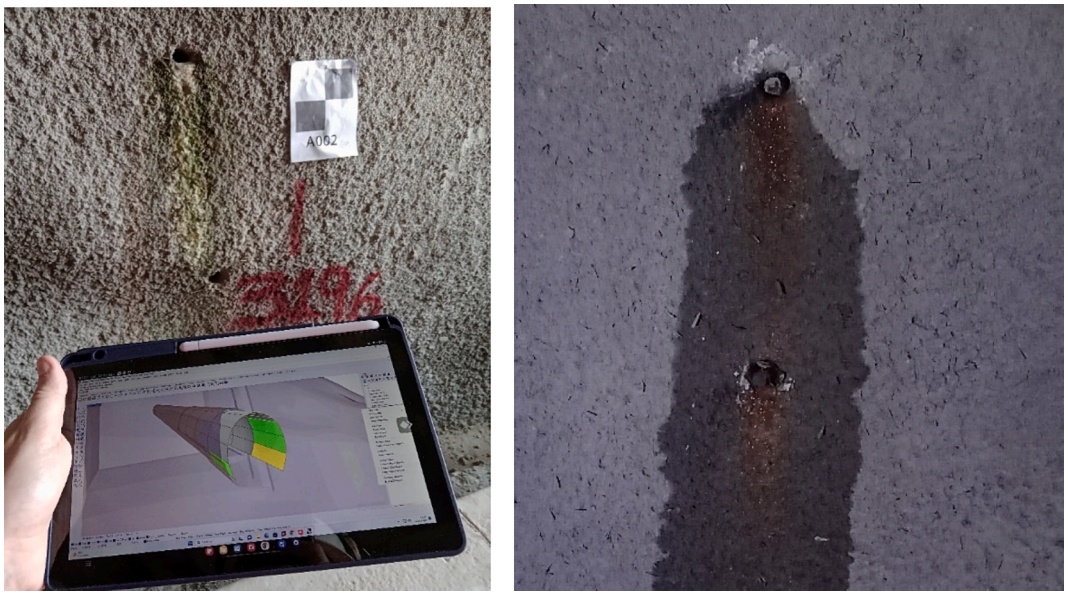
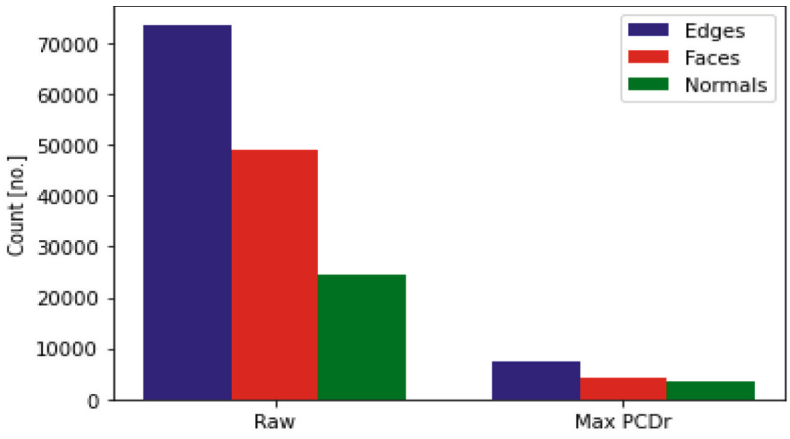
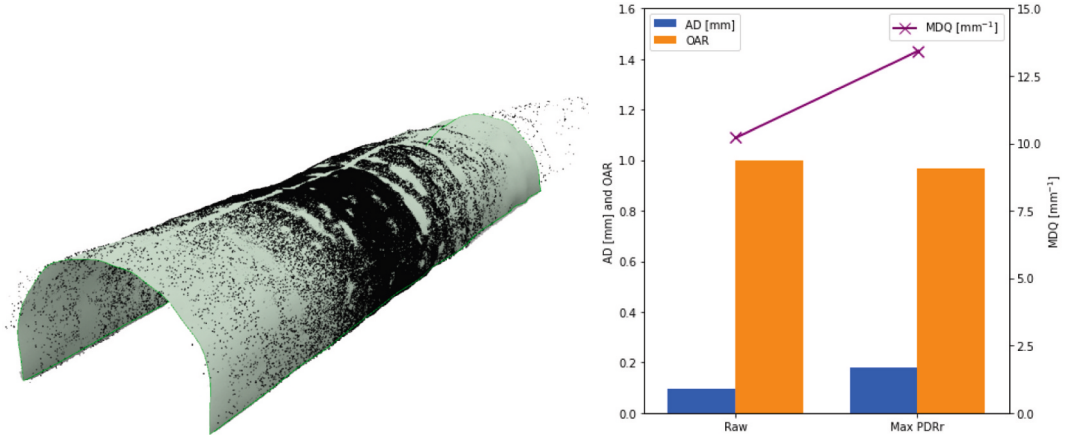


Fig. 15. Inspection data acquisition for generate the DT.



(a)



(b)

Fig. 16. Coastal tunnel procedural modeling application: (a) Decimate mesh properties; (b) mesh quality indexes and MDQ.

essential to obtain precise information for the process (Fig. 15).

Overexposure near the tunnel portal can result in washed-out colors or missing color information. To address this, several strategies can be employed: (i) pre-acquisition planning, such as scanning during periods of lower solar incidence (e.g., early morning or late afternoon) or using temporary shading structures to reduce glare; (ii) post-processing techniques, including color normalization and histogram equalization, although these require region-specific segmentation due to the high spatial variability in brightness; and (iii) the use of HDR-enabled scanners. In darker environments, although color capture may be limited, the scanner can still support defect identification through grayscale intensity values. When appropriately calibrated using a specific scale factor, these intensity readings can help detect features such as water leakage, surface cracking.

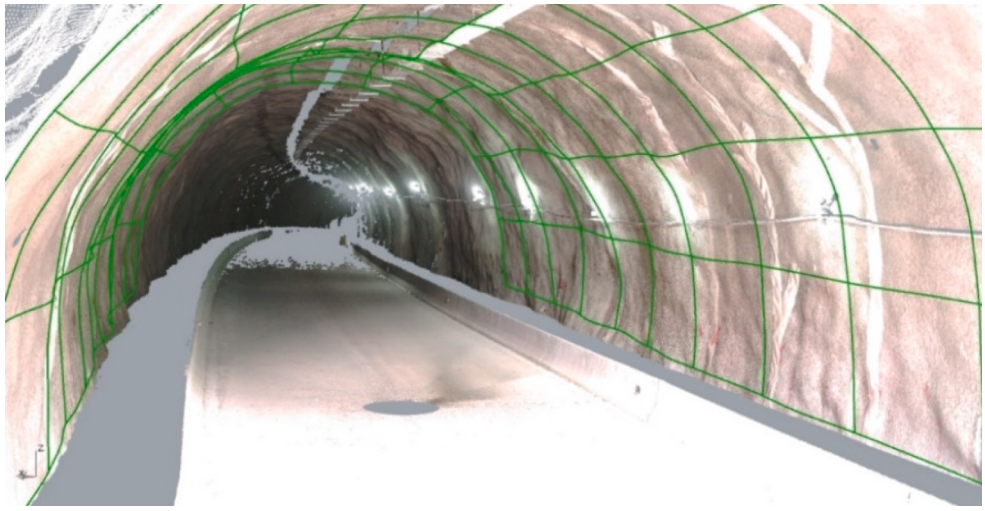
The inspection is conducted using surfaces generated overlaying the point cloud, with classification based on scanning information (Fig. 18). The TLS resolution level is a crucial parameter for the process feasibility, particularly in predominantly dark environments, where leak detection can be complex.

To test the digital inspection method, a 25-meter tunnel section was analyzed using both traditional visual inspection, performed by experienced infrastructure maintenance inspectors, and the proposed digital approach. The main objective was to validate and compare the limitations and improvements introduced by the procedural modeling process in tunnel inspection. For the degradation classification, the tunnel was divided into zones, allowing for a more detailed assessment of each method efficiency in detecting defects and accurately identifying their locations (Fig. 19).

The digital inspection produced results similar to those of the traditional visual (sensitive) approach. On the left wall, the visual inspection failed to detect and locate a leakage zone, likely due to the subjectivity of field interpretation and grid referencing. In contrast, the digital method provided a precise localization of the defect. The roof area, typically challenging for defect detection, also posed difficulties for the visual approach, which did not identify a constructive problem (C. P.). The analysis of the left wall highlighted an inefficiency in the proposed method regarding crack detection: while visual inspection successfully identified very fine shrinkage cracks, the digital method only detected larger cracks. However, this limitation is not critical, as the undetected cracks were small and not indicative of serious structural issues. Table 2 presents the defects and their locations identified by each approach.

The time consumption was evaluated (Fig. 20). The average duration of the traditional visual inspection was 46 min, while the digital inspection was divided into two stages: scanning time, used to capture geometry and images for generating the point cloud (approximately 10 min), and digital detection time (around 8 min), which included grid adjustment, mesh generation, and defect analysis. Overall, the digital inspection required less than half the time of the traditional method, while offering improved accuracy in defect identification and localization.

The Image inspection data is used to create a virtual image bank,



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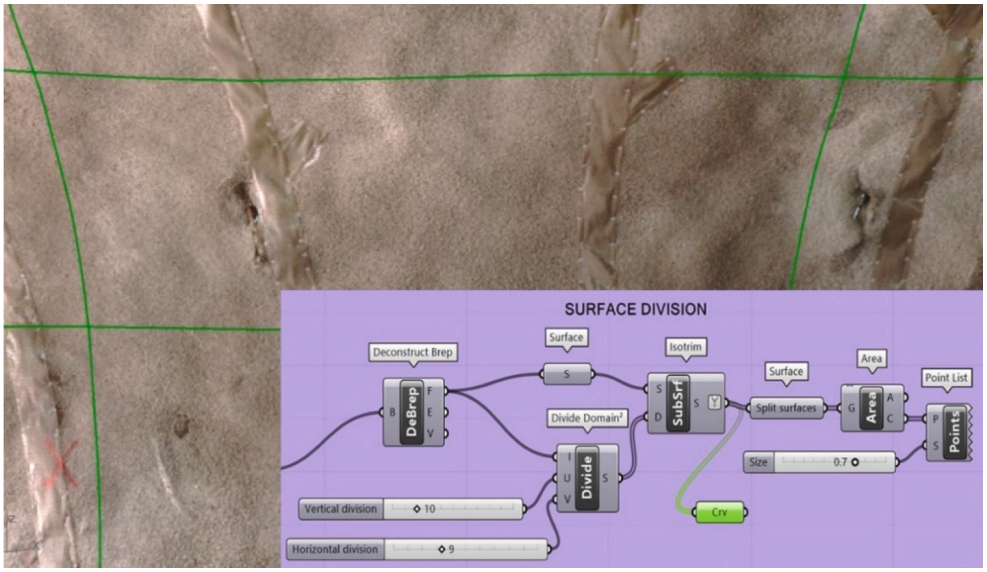

(b)

Fig. 17. Digital inspection development: a) Adaptable inspection grid built from the PCD; b) Digital inspection process and grid surface extraction.

with classification performed to generate the.csv file and inspection notes incorporated into metadata as key-value pairs for automatic assembly of the inspection grid information. The integration of this data and the inspection grid was achieved using EleFront 5.1.8, enabling the creation of a digital twin with automated structure reconstruction and real asset information connection, as shown in Fig. 21. The figure illustrates both common inspection campaign results and the procedural inspection DT approach. The results of defects recognition by digital inspection are presented in Table 3.

Table 3 shows the main defects identified using the framework for a 750-meter tunnel section. The results indicate significant leakage on the right wall and constructive problems (CP) in the roof, associated with shotcrete detachment. As a new infrastructure, this underscores the importance of conducting an initial inspection to identify susceptible zones for defects and potential deterioration areas. The results were compiled and are presented in an overview section.

4.2.4. Results overview

The programming development results span multiple stages, from the initial methodology design and validation using test tunnel PCD data to the creation of the digital twin and its application to the real Coastal Tunnel (Fig. 22). This research aims to establish a new inspection and data integration framework, connecting field assets with Industry 4.0 advancements for tunnel assessment.

Fig. 22 provides an overview of the development of each part of the programming and the main results of the digital twin at each stage in the construction environment. The main input data for the process are the point cloud data from the tunnel scanning and the inspection data from the sensitive inspection. Although the whole sensitive inspection is performed by standard digital modeling of the structure and the data is stored directly in the main server to mitigate errors in digital inspection, the result from dividing the surface using the grid and the point cloud data can serve as the validation of field acquisition data.

The inspection results from the proposed framework are summarized in Fig. 23. Fig. 23.a shows the grid inspection classification, presenting

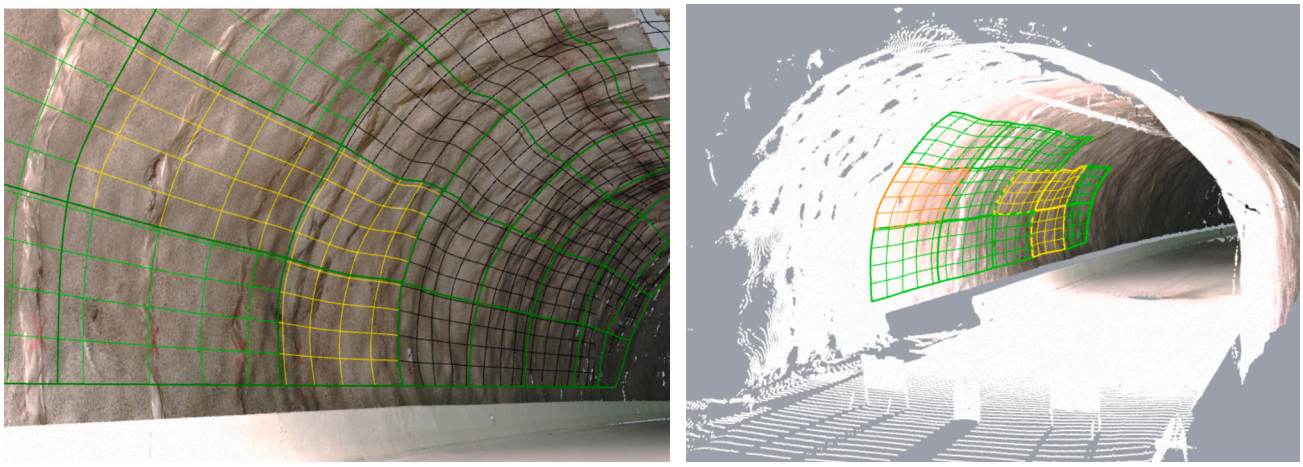

Fig. 18. Inspection digital classification process in the Coastal Tunnel.

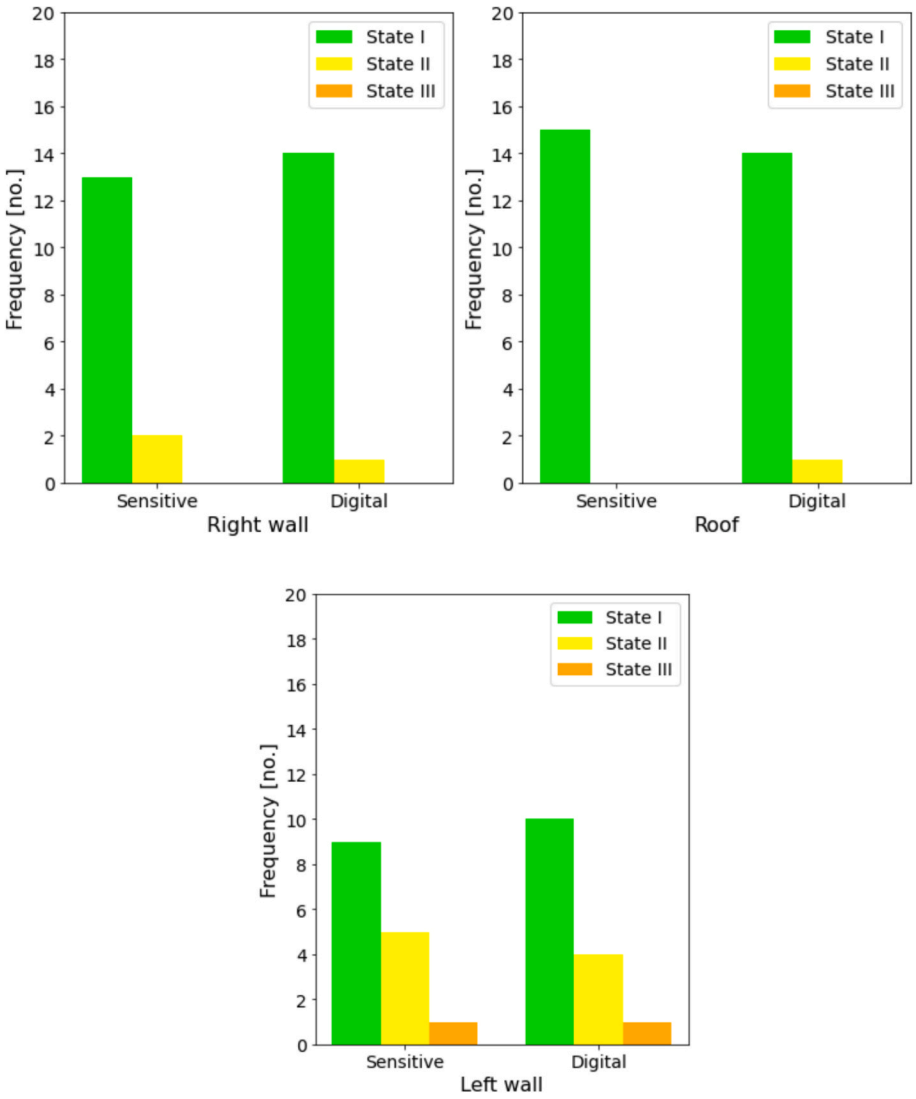

Fig. 19. Degradation state classification based on digital and sensitive inspections conducted during the validation test.

the service states of the structure. This inspection process, representing the cadastral or initial inspection, reveals a high number of cells in State I and none in State IV. This aligns with the initial defect identification, highlighting susceptible zones and optimizing the decision-making process. It also helps identify specific tunnel defects, improving the overall structure knowledge. Fig. 23.b shows that the main defects are leakage and CP, with a few points exhibiting cracking and one spalling area in the tunnel portal.

Table 2
Defects detection for each inspection approach.

Tunnel position [m]	Digital				Sensitive			
	Leakage	Cracking	Spalling	C.P.	Leakage	Cracking	Spalling	C.P.
2.5	1	0	1	0	1	1	1	0
7.5	0	0	0	1	0	0	0	0
12.5	0	0	0	2	0	0	0	0
17.5	2	0	0	1	1	0	0	1
22.5	0	0	0	0	0	0	0	0

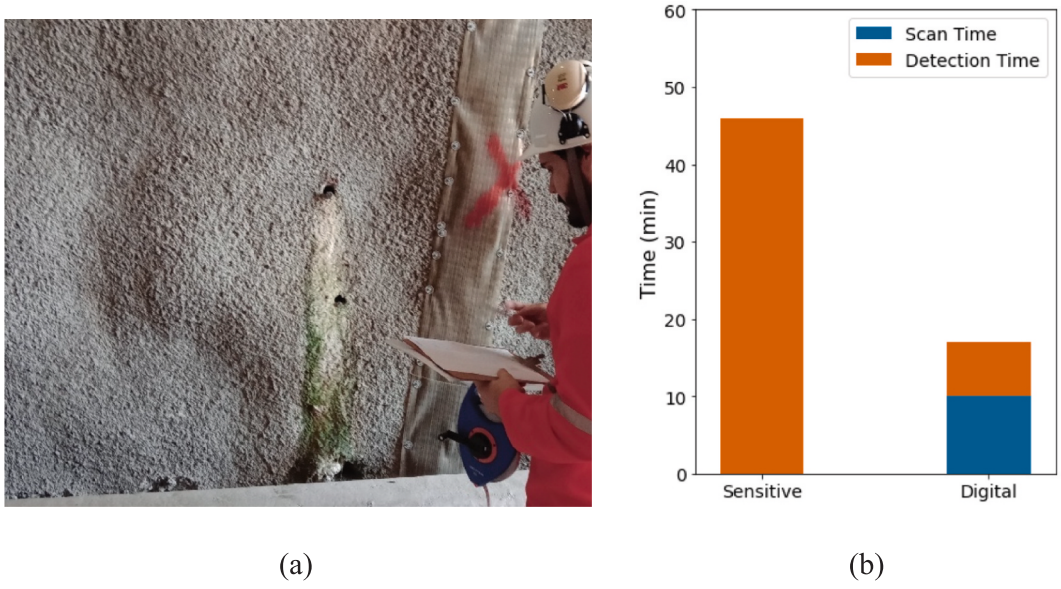

Fig. 20. Inspection validation: a) sensitive inspection; b) time consumption analyses.

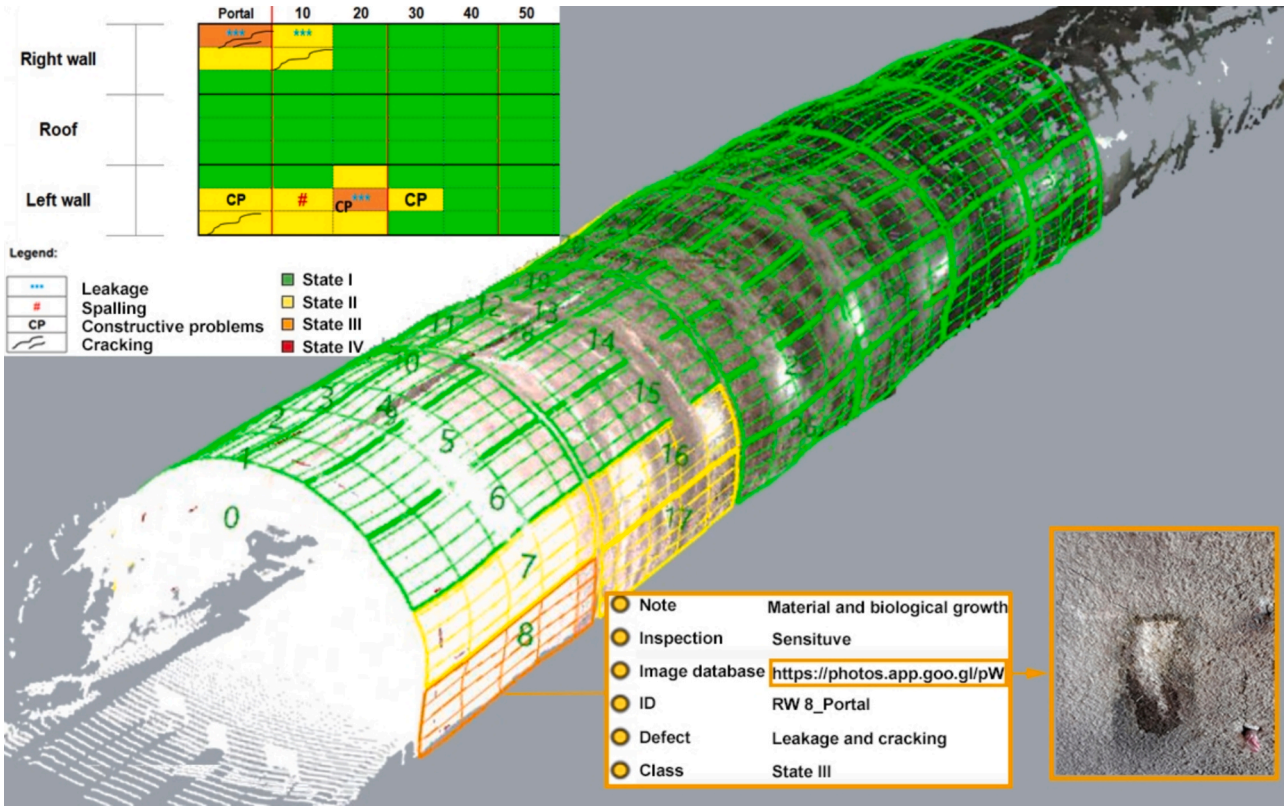

Fig. 21. Inspection digital twin for coastal tunnel.

Table 3
Digital inspection tunnel defects.

Tunnel	Inspection Defect			
	Leakage	Cracking	Spalling	Constructive Problems (C.P.)
Right wall	17	3	0	1
Roof	4	2	0	9
Left wall	3	1	1	3

The construction virtualization, combined with structural inspection information, represents a valuable resource for the management company, offering the possibility to enhance its decision-making regarding maintenance investments or more detailed inspection campaigns in areas of greater relevance. This process can be automated and continuously updated, integrating database files into online platforms, such as inspection classifications in.csv format, notes and information about the structure, and photographic images in online databases. The result of this process is the creation of a layer of information over time, which is adaptable to heterogeneous data from inspection campaigns for these types of structures.

4.2.5. *Advancements beyond existing methods*

The proposed framework introduces significant innovations compared to conventional inspection practices and previously reported Digital Twin (DT) approaches. It applies procedural modeling for the first time to tunnel DT lifecycle management, allowing flexible and parametric generation of inspection models and grids that adapt to diverse tunnel geometries. This enhances reconstruction accuracy, reduces occlusion effects, and significantly decreases processing time, supporting consistent updates throughout the asset lifespan.

A key contribution is the introduction of the Mesh Decimation

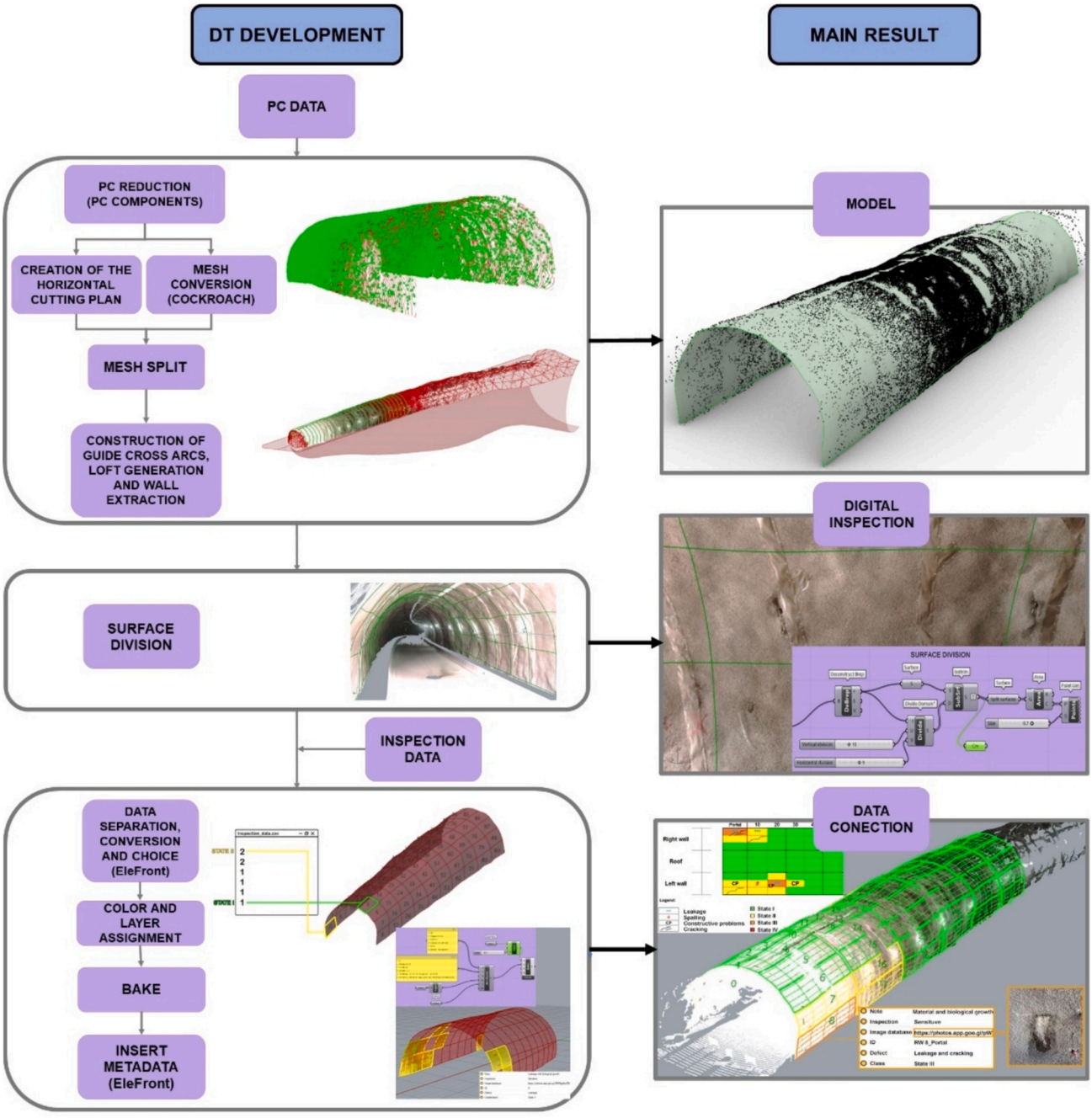

Fig. 22. Framework development and key results for each phase in the Coastal Tunnel.

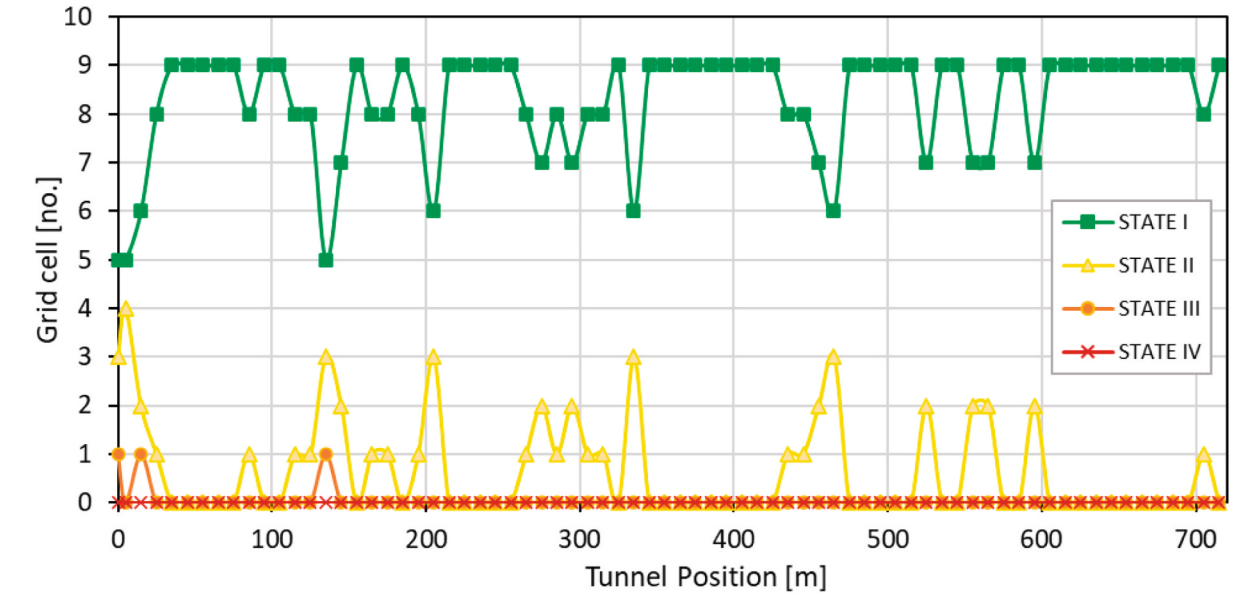

Fig. 23. Coastal tunnel digital inspection results: a) inspection grid classification; b) identified defects.

Quality (MDQ) index, a metric specifically designed for tunnel inspection applications. Unlike general-purpose mesh quality measures, MDQ integrates the Overlayed Area Ratio (OAR) and the Average Distance (AD) between vertices to capture both geometric fidelity and practical mesh usability (PCDr). This ensures that the resulting mesh remains computationally efficient while preserving the spatial precision needed for reliable inspection and modeling tasks.

Another distinguishing feature of the framework is the fully automated generation of inspection grids and metadata integration. In contrast to previous approaches that typically depend on semi-automated workflows or manual data handling, the proposed method directly embeds defect type, location, classification, and supplementary data into the digital model (Figs. 11 and 12). This tight connection between the physical tunnel and its virtual representation reinforces the digitalization twinning concept and enhances scalability and consistency, particularly in long-term monitoring and asset management scenarios.

5. Conclusion

The methodology developed for constructing a digital twin in tunnels, using parametric and procedural modeling, has proven effective in advancing the storage of virtualized information for managing and optimizing decision-making in infrastructure. The integration of new technologies and Industry 4.0 developments into assets of significant economic and social importance helps mitigate economic losses while

enhancing safety and longevity.

However, the methodology has limitations dependent on external factors or requiring adjustments during pre-processing. For instance, internet access is necessary to connect the inspection process to the main server. Point cloud density depends on scanning equipment and site characteristics, which can affect accuracy. Additionally, three-dimensional rotation adjustments may be required during pre-processing in the Fline feature, although these are relatively simple due to the flexibility of procedural modeling.

The MDQ index is introduced to ensure mesh workability and quality. This parameter is practical for guiding the meshing process: an 'acceptable' MDQ range indicates that the mesh resolution is sufficient to generate the tunnel mesh and grid without losing critical geometric information, while also optimizing computational cost. In contrast, a 'poor' MDQ may lead to inaccuracies in the model, potentially distorting the spatial allocation of inspection locations. These thresholds are crucial in ensuring that mesh quality meets the precision requirements necessary for reliable decision-making in digital tunnel inspection and modeling scenarios.

A digital twin demands a highly detailed model based on precise data from the real structure to minimize approximations. It is crucial for the twin to be linked to real asset information, which should be processed and integrated into the model efficiently. The digital replication of the Coastal Tunnel was performed using point cloud data from the Faro Focus 3D X330 TLS, and the proposed MDQ index proved effective for creating the mesh grid. Mesh construction was handled using Cockroach programming, leveraging the flexibility of procedural modeling to ensure a high LOD. Metadata integration from the real structure was automated using EleFront 5.1.8, and Grasshopper 1.0.0007 scripts with complementary open-source plugins facilitated efficient automation, detailing, and connection between real and virtual assets.

The framework developed for the Coastal Tunnel provides a digital inspection with a databank integrated into the tunnel model. The results meet digital twin requirements, delivering a virtualized, automated model connected to the asset for efficient management, identifying specific defects, improving the overall structure knowledge and maintenance plan optimization. There is potential for expanding the twin with additional layers of information, which will be explored in future research.

The results demonstrate the creation of automated geometric models from PCD data, virtualization of the inspection process, and metadata integration into the digital twin model. The framework offers flexibility and adaptability for various infrastructures and demands, enhancing tunnel lifecycle management and decision-making using Industry 4.0 technologies.

CRediT authorship contribution statement

Lucas Bellini Machado: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Bernardo Lopes Poncetti:** Writing – review & editing, Writing – original draft, Validation. **Leandro Silva de Assis:** Writing – review & editing, Validation. **Marcos Massao Futai:** Writing – review & editing, Supervision, Resources, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix

Table A1
Inspection country specific standards.

Classification System	Classification Basis	Levels of Deterioration
FHWA (2015)	Structural components based on characteristic units of measurement	4 levels
CS 452 (2020)	General classification of tunnel state	5 levels
TSI (Li et al., 2017)	Based on the serviceability of the tunnel	5 levels
ADIF (2021)	Severity related to the type of damage, affected element, and intensity	4 levels
CeTU (2015)	Civil engineering and water components classified separately	5 levels (civil) and 3 levels (water)

Data availability

Data will be made available on request.

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